

SHIVAM GUPTA

AI Engineer | LLM & GenAI Systems

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🌐 [My Portfolio](#)

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Summary

AI Engineer with 2 years of experience building production **LLM, RAG, and voice-AI systems** using **Python, FastAPI, Redis, MongoDB** and the **Gemini, OpenAI, and Anthropic** APIs. Core contributor to a real-time conversational voice platform taken from first prototype to production with real clients (response latency **7–8 s** → **under 900 ms**); tuned RAG retrieval to **recall@5 of 0.90** on hard queries; **4 research papers** (IEEE, Springer; 1 in press).

Skills

Languages: Python, JavaScript, C/C++, SQL, JSON

LLM / GenAI: LLM agents & function calling (Gemini, OpenAI, Anthropic APIs), RAG, prompt engineering, hybrid retrieval (BM25 + dense, RRF), cross-encoder reranking, embeddings (sentence-transformers), retrieval evals (recall@k, MRR), vector stores (Chroma, OpenSearch)

ML Frameworks: TensorFlow, PyTorch, scikit-learn, OpenCV, Pandas, NumPy

Backend & Infra: System design, FastAPI, Flask, Django, REST APIs, WebSockets, SSE streaming, Redis, MongoDB, MySQL, SQLite, GCP

Testing & Tools: Playwright, pytest, Git/GitHub, LaTeX, VS Code, Jupyter

Experience

BELZABAR SOFTWARE PRIVATE LIMITED

Sep 2024 – Present

Computer Scientist (AI Engineer) | Jul 2025 – Present

New Delhi, India

- Core contributor to the **ground-up development** of the **Expertly Voice** platform, a production real-time conversational AI voice-call system now serving **real clients** and supporting **100 concurrent calls**.
- Contributed to the **end-to-end architecture** across multiple redesigns, absorbing new client requirements without major rewrites.
- Cut voice response latency from **7–8 s to under 900 ms** by optimizing the pipeline, introducing **Redis** caching for redundant work, and tuning **MongoDB** queries for user-specific data fetching.
- Shipped **five advanced call capabilities**: mid-call language switching, behavior handoff between multiple AI roles, adding another person to the call, live **tool execution**, and modifying call behavior mid-call without the caller noticing.
- Designed an **extensible architecture** with documentation and implementation guides that let new capabilities and **third-party integrations** ship without reworking the core system, reducing integration effort for every feature that followed; implemented the **permission module**.
- Optimized the **Expertly Knowledge RAG ingestion pipeline** by batching LLM similarity-scoring calls (titles: 20 calls → 1; snippets: 20 calls → 5), cutting ingestion time of long documents by **~40%** and scoring-stage token cost by **~40%** while preserving ranking quality.
- Presented architecture decisions, trade-offs, and optimization results to stakeholders and clients in **weekly demos** and recurring client meetings.
- Mentor an intern and guide new team members through their initial phase on the team.

AI Engineer (Intern)

Sep 2024 – Jun 2025

- Developed **custom AI agents** in **Python** using large language models, automating domain-specific workflows tailored to client requirements.
- Customized pre-trained **LLM-based** solutions to interpret complex queries and execute focused domain tasks; integrated and maintained backend services across multiple client implementations.

EZEIATECH SYSTEMS PVT. LTD.

Jan 2023 – Mar 2023

Machine Learning Intern

Gurugram, Delhi NCR

- Led the development of an AI model for unit tests on **5,000+** data points, achieving a **10% improvement** in throughput.
- Automated test-result reporting with Python and TensorFlow scripts, reducing reporting time by **20%**.

Projects

Jarvis — Voice-Driven Agentic AI Assistant | *Python, LLM function calling (Gemini/Claude/OpenAI), Vosk, Edge-TTS*

- Built an end-to-end agentic voice assistant for Windows: offline wake-word and speech-to-text (openWakeWord, Vosk) driving cloud LLMs with **multi-round function calling across 13 guarded tools**, delivering spoken replies via Edge-TTS.
- Engineered a **provider-neutral LLM layer** that unifies three incompatible function-calling protocols (Gemini, Claude, OpenAI) behind one interface, so swapping providers takes a **single configuration change**.
- Treated model output as **untrusted input**, enforcing an adversarially tested filesystem sandbox and strict argument allowlists, all verified by a **~290-check automated eval harness**.

Synapse — Retrieval-Augmented Generation System | *Python, FastAPI, Chroma, sentence-transformers, BM25, Gemini*

- Architected a document-chat RAG system with a **custom retrieval core** (no LangChain/LlamaIndex), serving grounded, **source-cited answers** with an honest-refusal gate that prevents hallucination on out-of-corpus questions.
- Raised hard-query **recall@5 from 0.40 → 0.90** via **hybrid retrieval** (dense + BM25, Reciprocal Rank Fusion) and **cross-encoder reranking**, tuned against an offline eval harness (recall@k, MRR).
- Shipped a 4-screen FastAPI web app (chat, knowledge-base management, tuning sandbox, analytics) with a **pluggable LLM layer** running Gemini in production and OpenAI/Anthropic behind the same interface.

- Lens — AI Reading Assistant (Chrome MV3 Extension)** | *JavaScript, SSE streaming, Gemini/OpenAI/Anthropic, Playwright*
- Created a privacy-first, **zero-dependency** Chrome side panel, **published on the Chrome Web Store**, that streams AI chat about any web page with no backend and user-owned API keys, unifying three providers’ SSE streaming dialects behind **one adapter**.
 - Reverse-engineered client-side extraction of **canvas-rendered Google Docs and Sheets** from Google’s obfuscated in-memory document models, without Google APIs or OAuth.
 - Delivered multi-tab AI context with **zero added Chrome permissions**, an XSS-safe streaming markdown renderer, and a **107-case Playwright** end-to-end suite running in real Chromium.

Publications

- “Gesture Recognition in Sign Language Translation: A Deep Learning Approach” – **IEEE** (2024)
- “Developing Game AI for Classic Games Using Reinforcement Learning” – **IEEE** (2024)
- “Deep Learning in Structure From Motion” – **Springer** (2026)
- “Early Detection of Diabetic Retinopathy Using SMOTE Technique Analysis of Retinal Images” – **Springer** (presented; in press) — IndiaAI Fellowship

Education

Chandigarh University
*Bachelor of Engineering in Computer Science, specialization in Artificial Intelligence — **CGPA: 8.63***

Aug. 2021 – May 2025
Punjab, India

Certifications

Google Data Analytics Professional Certificate - Google	2024
Applied Machine Learning in Python - University of Michigan	2024
Introduction to Data Science - IBM	2023
Analytics for Decision Making - Coursera	2023

Awards and Recognition

- **IndiaAI Fellowship:** Year-long AI research fellowship; research on early detection of diabetic retinopathy
- **Academic Excellence Award:** Chandigarh University
- **SIH’23 Finalist:** Grand finalist, Smart India Hackathon 2023
- **Ranked 1st:** Inter-university Hackathon by Google Developer Club and Devfolio
- **Ranked 2nd:** All India ELECTROTHON 5.0 by MLH and GitHub (NITH)